



C1 Report

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Business Intelligence

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1. Introduction and evolution from W1

This report develops the commercial alternative selected in W1 using the DataCo Supply Chain dataset. The selected line examines sales behavior across regions and product categories, with the main question: How do the best-selling product categories vary across regions, and what factors are associated with lower sales volumes?

The proposed user is a commercial analyst. The proposed decision is which region-category combinations should receive further commercial investigation before making assortment changes or pursuing promotional opportunities. The analysis is historical and exploratory; it supports prioritization for further investigation rather than direct prescriptions.

1.1 W1 alternatives and selection

Alternative	Objective	Available evidence	Future continuity
Candidate A - Sales	Analyze differences in sales behavior across regions and categories.	Region, category, quantity, sales, price, discounts and customer variables.	Sales prediction; commercial dashboard.
Candidate B - Logistics	Analyze late-delivery risk by region, category and shipping mode.	Late_delivery_risk, region, category, shipping mode and delivery times.	Late-delivery risk classification; logistics dashboard.

Candidate A was retained because region, category, quantities, prices and discounts support the proposed commercial comparisons. Candidate B remains feasible, but its interpretation is limited by missing operational causes and by the need to verify whether potential predictors are known before delivery.

The notebook also retains the W1 logistics evidence: item-weighted late-delivery-risk shares range from 48.80% in Canada to 57.96% in Central Africa, while the four shipping modes show different mean values. These results were used as supporting evidence for comparing the alternatives rather than as the focus of C1.

1.2 Documentary foundation

The commercial line is supported in the notebook by Trivedi (2011), which motivates combined geographic and product-category analysis, and by Leeflang and Parreño-Selva (2012), which supports considering category, price and promotion variables. The logistics alternative is supported by Niemi et al. (2020) and Harter, Stich, and Spann (2025), which motivate attention to delivery punctuality and its potential commercial consequences. The notebook explicitly treats observed relationships as associations rather than causal effects.

1.3 Response to instructor feedback

According to the notebook, the C1 revision responded to feedback by adding two academic sources for Candidate A, explicitly defining the observation unit as an Order Item, stating the covered period, executing and saving the notebook with outputs, and strengthening the reproducibility documentation.

2. Seven-step analytical framework

Step	Application to the selected commercial line
1. Decision	A commercial analyst prioritizes region-category combinations for further investigation. Alternatives are volume-led prioritization,

	investigation of low-volume groups, or no action pending stronger evidence.
2. Success criteria	Produce defensible comparisons with explicit population, denominators and limitations. A future increase in sales is not measured as an intervention effect.
3. Business questions	Compare category shares and units per order across regions; explore item quantity by discount band within categories.
4. Business process	Order registration -> item quantities/prices/discounts -> recorded order state -> shipment. Inventory, offer exposure, competitor prices, returns/refunds and decision reasons are not established.
5. Entities, events and states	Customers place orders containing items. Order state is a recorded status, not proof of payment or realized revenue. A non-purchase is not an observed item row.
6. Data representation	One row is one order item; Order Item Id is the candidate key; Order Id connects multiple items. One supplied source is used and no external join is required.
7. Measurement	Formulas are applied to a stated population and period. The exploratory commercial subset is compared with the all-record baseline.

2.1 Measurement definitions

- Recorded units: sum of Order Item Quantity over eligible items.
- Distinct orders: number of distinct Order Id values represented in the same eligible population.
- Units per order: recorded units divided by distinct represented orders.
- Region-category share: eligible units of a category in a region divided by all eligible units in that region, multiplied by 100.
- Recorded sales amount per order: sum(Sales) divided by distinct represented orders; this is not treated as realized revenue or profit.
- Mean quantity per item in a discount band: sum(quantity) divided by the number of eligible items in that category and band.
- Empty groups are treated as missing or unobserved, not automatically as zero demand.

The notebook states an initial analytical window from January 1, 2015 to January 31, 2018, with the actual coverage recalculated from the data. January 2018 is not treated as a complete year, and no direct policy match is claimed for this commercial dataset.

3. Data source, profiling and preparation

The DataCo Supply Chain dataset contains 180,519 rows and 53 variables. The notebook describes the original CSV as the source file and also describes a Parquet copy as an equivalent compressed representation. In the executed notebook supplied for this report, the loading cell reports the actual input as '../data/dataset.parquet'.

Profile check	Executed result
Records	180,519
Variables	53
Unique Order Item Id	180,519
Unique Order Id	65,752
Unique customers	20,652
Categories	50
Regions	23

First recorded order date	2015-01-01 00:00:00
Last recorded order date	2018-01-31 23:38:00
Duplicate rows	0

The dataset-level missing-value inspection reports complete values for the fields used in the commercial analysis. Other dataset fields do contain missing values: Product Description is missing in all 180,519 rows, Order Zipcode in 155,679 rows, Customer Lname in 8 rows and Customer Zipcode in 3 rows. These fields are not used as essential variables in the selected analysis.

Essential field	Check
Order Region	0 missing
Category Name	0 missing
Order Item Quantity	0 missing
Order Item Discount	0 missing
Order Status	0 missing
Order Item Id	0 missing; 0 duplicates
Order Id	0 missing

3.1 Observation unit and temporal coverage

The unit of observation is an Order Item. Each row is a product line associated with an order and includes product, quantity, sales, customer, regional and logistics information. Therefore, the 180,519 rows do not represent 180,519 independent customers or orders.

Coverage is materially uneven. The notebook counts only six months with records for Canada, Central Africa, Central Asia, East Africa, Eastern Europe, North Africa, Southern Africa, West Africa and West Asia, while Northern Europe, Southern Europe and Western Europe have records in 18 months. Category coverage also differs substantially across time.

3.2 Commercial subset

Order Status	Items
COMPLETE	59,491
PENDING_PAYMENT	39,832
PROCESSING	21,902
PENDING	20,227
CLOSED	19,616
ON_HOLD	9,804
SUSPECTED_FRAUD	4,062
CANCELED	3,692
PAYMENT_REVIEW	1,893

For the exploratory commercial subset, the notebook excludes 3,692 CANCELED items and 4,062 SUSPECTED_FRAUD items. No essential-field rows are invalid. This leaves 172,765 retained items representing 62,897 distinct orders. The exclusion is an explicit analytical assumption; it is not evidence that all remaining records are completed sales.

4. Data quality - Five Cs

Lens	Application and decision
Clean	Identifiers, numeric validity, dates, statuses and monetary

	arithmetic are checked. Invalid essential fields would be excluded; CANCELED and SUSPECTED_FRAUD are excluded from the commercial subset.
Consistent	Monthly regional coverage and within-order date/region consistency are checked. The analysis does not assume stable category availability or reporting intensity.
Conformed	No external source is joined. Cross-source harmonization is therefore not applicable; the internal Category Id/Category Name mapping is validated.
Current	The data describe 2015-2018. They are appropriate for a historical educational analysis but not for immediate operational decisions in 2026.
Comprehensive	Region/category/month coverage and sample sizes are inspected. Inventory, promotion exposure, unmet demand and complete lifecycle histories are not established.

The executed quality checks found zero conflicting region or date values within retained orders and no Category Id mapped to multiple Category Name values. Monetary arithmetic checks found no discrepancies above 0.05 for Sales versus price x quantity or for Order Item Total versus Sales minus Discount in the retained subset. The notebook performs no imputation, no external join and no raw-file overwrite.

5. Exploratory commercial analysis

The main C1 question is: within the recorded DataCo transactions from January 2015 to January 2018, how do category shares and units per order differ across regions, and how does recorded quantity per item vary with discounts within comparable categories?

5.1 All-record baseline

Order Region	Units sold	Item records
Central America	62,091	28,341
Western Europe	56,504	27,109
South America	32,647	14,935
Oceania	20,430	10,148
Northern Europe	20,402	9,792
Canada	2,183	959
Central Asia	1,214	553

In the all-record baseline, Central America has the largest recorded quantity, followed by Western Europe and South America. The notebook emphasizes that absolute values should not be interpreted automatically as regional preferences because regions with more represented orders will tend to accumulate more units.

Top category	Recorded units (all records)
Cleats	73,734
Women's Apparel	62,956
Indoor/Outdoor Games	57,803
Cardio Equipment	37,587
Shop By Sport	32,726

5.2 Quality-aware regional comparison

Region	Retained units	Orders	Items	Units / order
Central America	59,540	9,001	27,174	6.61
Western Europe	53,943	9,563	25,867	5.64
South America	30,978	4,738	14,184	6.54
US Center	12,471	1,852	5,653	6.73
Eastern Europe	8,470	1,246	3,785	6.80
Canada	2,063	291	907	7.09
Central Asia	1,190	180	542	6.61

The retained commercial subset shows why volume and purchase intensity should be read separately. Central America records 59,540 retained units, while Canada records only 2,063 retained units but the highest units per represented order among the displayed results, at 7.09. Canada is based on only 291 represented orders, so the intensity estimate must be read together with sample size and coverage.

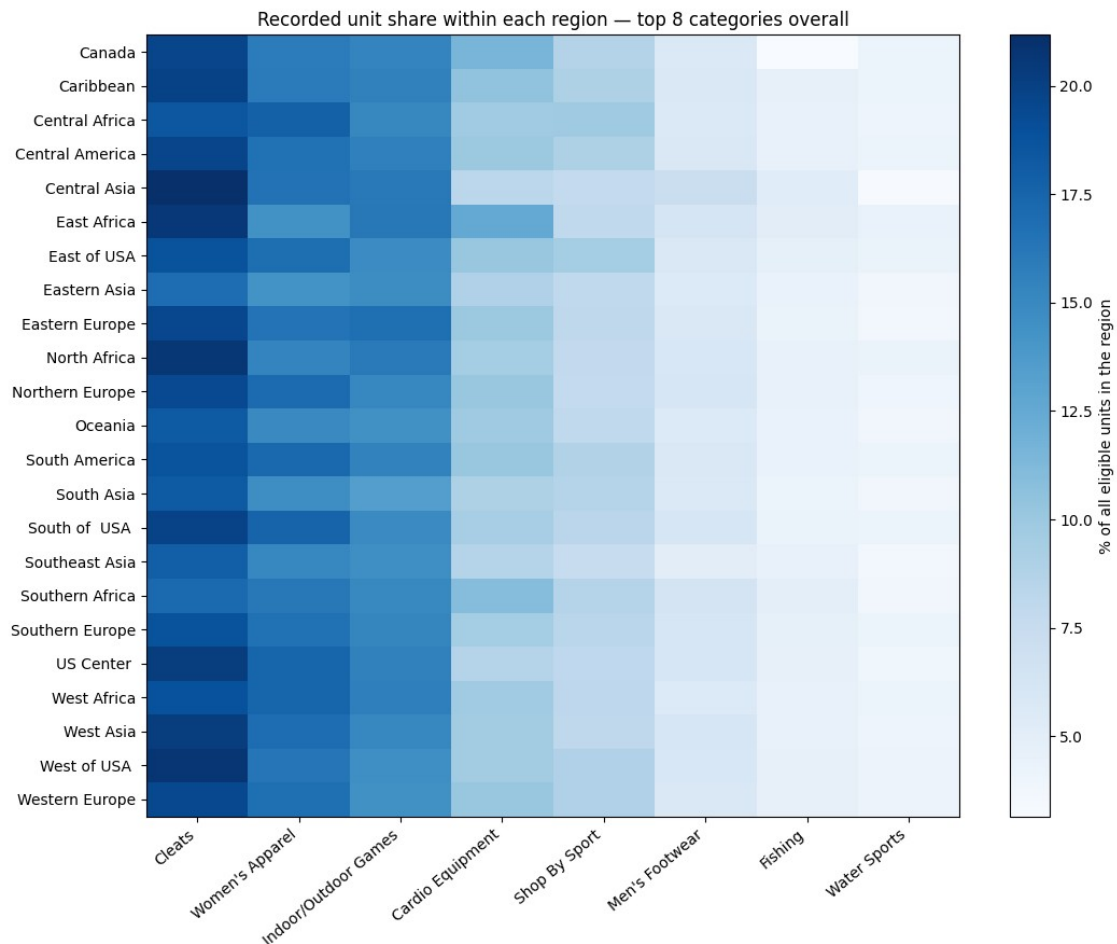


Figure 1. Category composition across regions using the retained commercial subset. Missing combinations are unobserved, not verified zero demand.

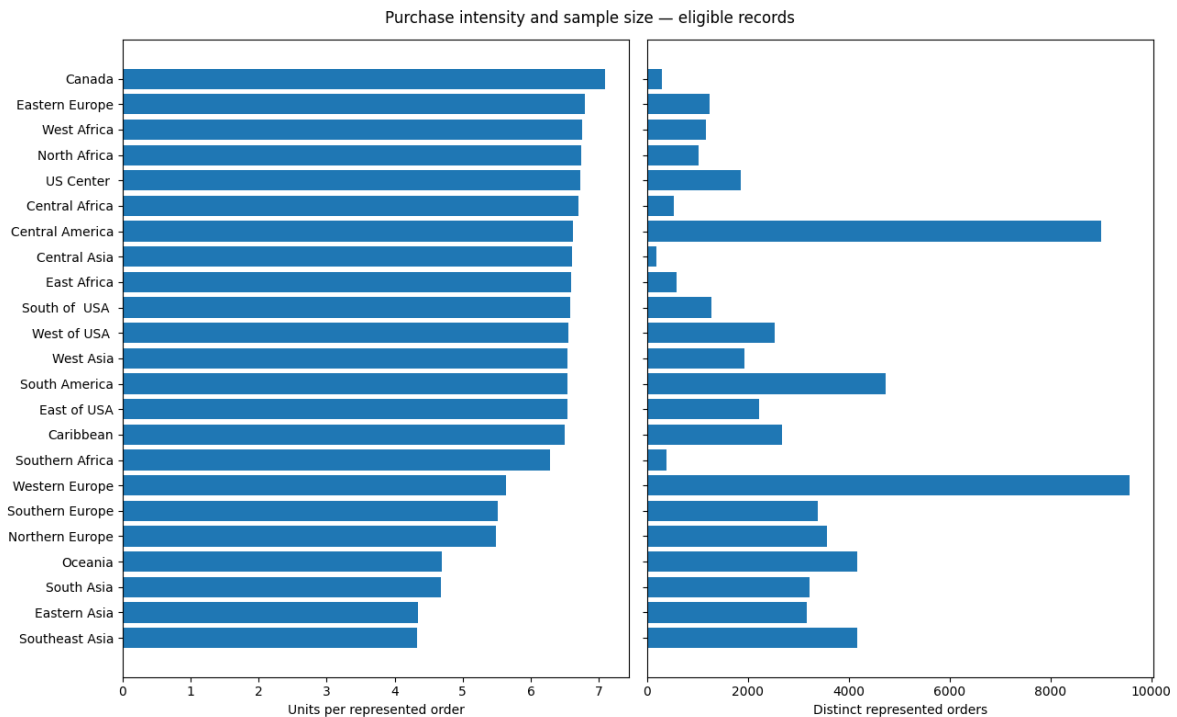


Figure 2. Regional purchasing intensity and represented-order sample size. Volume and units per order answer different questions.

5.3 Sensitivity to status exclusions

Largest observed share changes	Change after exclusions
Southern Africa - Women's Apparel	-0.542 pp
Central Asia - Indoor/Outdoor Games	-0.506 pp
Central Asia - Cleats	+0.419 pp
Canada - Accessories	-0.265 pp
Central Asia - Cardio Equipment	-0.251 pp

The largest observed absolute change in region-category share after excluding CANCELED and SUSPECTED_FRAUD items is approximately 0.54 percentage points. This indicates that the main composition results are relatively stable to this specific exclusion rule, but it does not establish realized revenue or validate every remaining order state.

5.4 Discount comparison within categories

Category	Max - min mean quantity across discount bands
Cleats	0.008
Women's Apparel	0.009
Indoor/Outdoor Games	0.039
Cardio Equipment	0.015
Shop By Sport	0.045

Across the five largest categories, the mean quantity per recorded item remains close to three units across discount bands. The difference between the highest and lowest band-specific mean is only 0.008-0.045 units. The notebook therefore does not support a recommendation for deeper discounts. Customer selection, product mix, region, time and price are not held constant, and no elasticity, significance test or causal effect is estimated.

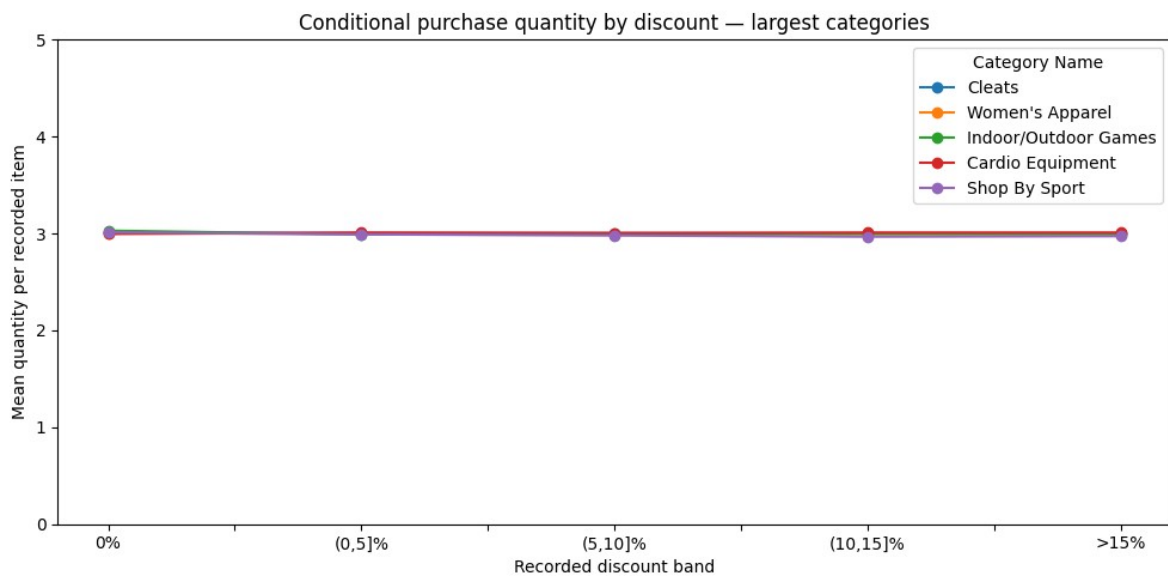


Figure 3. Mean quantity per recorded item by discount band for the five largest categories. The comparison is descriptive and conditional on an item being recorded.

5.5 Main comparability limitation

Temporal coverage is the most important limitation for regional interpretation. The notebook reports that Canada's observations occur from August 2016 through January 2017, while other regions have different observation windows. Product categories also have unequal date coverage. Whole-period regional rankings therefore combine time, assortment and geography. Units per order reduce scale differences but do not remove these confounders.

A subsequent comparison should restrict the data to overlapping periods and available categories and should report the resulting restricted sample. Lower recorded sales should not be interpreted as proof of market failure or low underlying demand.

6. Conclusions and decision implications

- The source contains 180,519 unique item records. The retained commercial subset contains 172,765 items representing 62,897 orders.
- Central America has the largest retained recorded volume among regions in the notebook, with 59,540 units.
- Canada has 2,063 retained units and 7.09 units per represented order, but its sample and temporal coverage are much smaller.
- The maximum observed category-share change after excluding CANCELED and SUSPECTED_FRAUD items is about 0.54 percentage points.
- Across the five largest categories, discount-band differences in mean item quantity are small (0.008-0.045 units).
- Whole-period regional rankings are descriptions of the recorded DataCo data, not directly comparable measures of regional performance.

The proposed decision is to use recorded volume, category shares, units per order and sample sizes jointly to shortlist region-category combinations for deeper investigation. The notebook explicitly recommends reviewing reporting coverage and assortment first and then seeking inventory,

promotion exposure/cost, margins and customer-opportunity data before making investment decisions.

Alternative explanations include regional activity levels, product availability, category mix, order states and unequal temporal coverage. Currency and accounting meaning must also be verified before interpreting monetary variables as realized revenue.

7. Future ML and dashboard feasibility

7.1 Possible future ML task

A possible future task is forecasting next-month recorded units for a region-category pair. The unit would be region-category-month and the outcome would be next-month units. Candidate inputs include prior-month quantities, prior order counts, lagged category shares and calendar variables.

Realized prices, discounts, order states and sales from the forecast month are not available at prediction time and should not be treated as known predictors. The approximately three-year window, incomplete final year and uneven category coverage may make many series unsuitable. If contiguous comparable series are unavailable, the notebook proposes exploratory clustering of region-category profiles within overlapping periods as a fallback. No model is trained and no performance is promised in C1.

7.2 Proposed dashboard

The proposed user is a commercial analyst or category manager. Core indicators are recorded units, represented orders, units per order and within-region category shares. Suggested filters include month, region, category, customer segment and inclusion/exclusion of flagged statuses. The dashboard should display sample sizes and coverage warnings alongside rankings and should separate composition and purchasing intensity from volume.

8. Process record and reproducibility

8.1 Execution and environment

Component	Executed version
OS platform	Linux-7.2.7-200.fc44.x86_64-x86_64-with-glibc2.43
Python	3.13.2
pandas	2.3.2
NumPy	2.3.2
Matplotlib	3.10.5

The notebook instructs the reviewer to execute it from a fresh kernel, top to bottom, using project-relative paths and to save all relevant outputs. The project repository listed in the notebook is <https://github.com/ians513/Business-Intelligence>. The repository is supplementary; the final package must remain self-contained.

A reproducibility detail remains visible in the executed notebook: although the process record states that the C1 revision would load the CSV directly instead of an optional Parquet cache, the actual loading output for the supplied execution reports `'../data/dataset.parquet'`. The notebook describes that Parquet file as an equivalent compressed copy of the original CSV. The final package should keep the executed input and documentation aligned.

8.2 Decisions and checks

Stage	Decision	Reason / verification	Notebook evidence
W1	Retain sales rather than logistics	Compared available fields, limitations and potential uses.	Sections 1 and 6
C1 revision	Preserve all-record summaries and add explicit eligibility rules	Inspected status counts and compared shares before/after exclusions.	Sections 4 and 5
C1 revision	Use category shares and distinct-order denominators	Avoid confusing regional sample size with purchasing intensity.	Sections 2 and 5
C1 revision	Describe associations, not causal effects	Considered coverage, product mix, availability and purchase selection.	Sections 5 and 6

8.3 Individual contributions

Ian Spikin Thomas contributed to the definition of the problem, data preparation, exploratory analysis, interpretation of results and preparation of the final deliverables.

Juan Pablo Molina participated in the project framing, data review, analytical development, discussion of findings and completion of the report, notebook and presentation.

Both members worked collaboratively throughout the project and contributed at a similar level.

8.4 AI use and verification

The notebook records that ChatGPT/Codex assisted with source searching, bibliography organization, rubric comparison, code additions and editorial restructuring. Relevant requests included support implementing the process for Candidate A, organizing title/link/description information for Candidate B, and reviewing/correcting the notebook while preserving the team's work.

Source suggestions were modified by the team. The final commercial references were Trivedi and Leeflang/Parreño-Selva, and an earlier source suggestion was replaced. Verification is evidenced by source links, saved execution, field/range checks, before/after counts and arithmetic tests. Execution verifies calculations, not causal interpretations.

8.5 Individual reflection

Ian Spikin Thomas: This activity reinforced the importance of clearly defining the unit of observation, understanding the limits of the available data, performing an EDA, applying the Five Cs, and understanding the stages of a data-analysis project.

Juan Pablo Molina: The project provided experience with the full data-problem-solving pipeline and with structured techniques such as the Five Cs and the seven-step analytical framework. The regional sales analysis also showed how patterns and differences emerge when sales data are broken down across dimensions.

9. References

Constante, F., Silva, F., & Pereira, A. (2019). DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS, version 5. <https://doi.org/10.17632/8gx2fvg2k6.5>

Trivedi (2011). Regional and Categorical Patterns in Consumer Behavior: Revealing Trends.

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Harter, Stich, and Spann (2025). The Effect of Delivery Time on Repurchase Behavior in Quick Commerce.

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